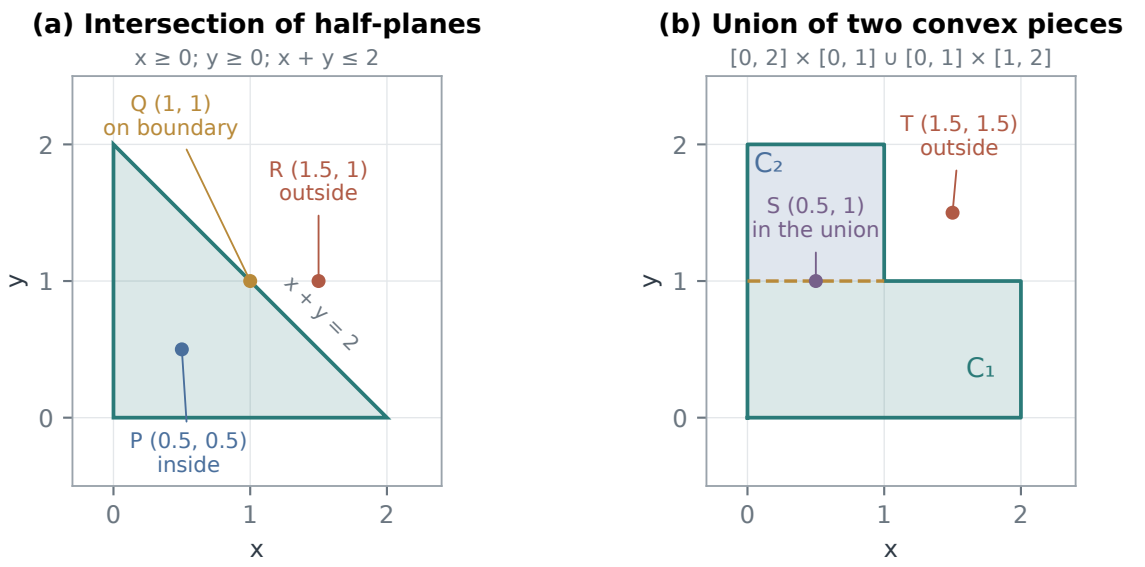

P: INSIDE; Q: BOUNDARY; R: OUTSIDE
Illustrative classification with ε = 0.

The dashed line separates the two pieces.
S is not on the outer boundary of the union.